

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show / Hide Password Visibility Toggle

Student Management System

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle
Priority	Medium
Change Type	Frontend Enhancement / Usability and Accessibility
Affected Modules	Login and Student Management (Add Student form)
Implementation Status	Code updated; testing and screenshot evidence to be verified
Date	8 October 2026
Prepared By	Y. Ugatharsini
Student ID	JFICT24_33

1. Change Request Summary

1.1 Business Requirement

Users and administrators enter passwords into masked fields and may make typing errors without being able to verify the entered text. The requested change adds a clear show/hide control so users can check a password before submitting the form. An additional matching control was added to the Add Student password field for consistent usability.

1.2 Existing Behaviour

The LoginFormSection component used a password input with type="password" at all times and did not provide a visibility toggle. The StudentFormModal component also masked the password entered when creating a new student, with no option to reveal or hide it manually.

1.3 Requested and Additional Change

- Add a show/hide password button to the login password field.
- Switch the input type between password and text based on component state.
- Show an eye icon while the password is hidden and an eye-slash icon while it is visible.
- Start with the password hidden whenever the component/page is opened.
- Use a type="button" toggle to avoid accidentally submitting the form.
- Provide aria-label and aria-pressed attributes and retain native keyboard operation.
- Adjust input padding and button positioning so the password text does not overlap the icon.
- Apply the same visibility toggle to the Add Student password field in StudentFormModal.jsx.
- Preserve existing validation, error handling, loading behaviour, API requests, authentication and role-based navigation.

1.4 Scope

In Scope	Out of Scope
Login password visibility toggle; Add Student password visibility toggle; client-side visibility state; icon and input styling; keyboard and screen-reader accessibility; responsive layout; preservation of existing submission flows.	Database/schema changes; backend controller/model changes; new API endpoints; JWT or authentication changes; password hashing changes; password reset workflow; changes to student CRUD rules or dashboard logic.

2. Implementation

2.1 Implementation Summary

The change uses React useState to control whether each password input is masked or visible. The input type is selected dynamically: password when visibility is off and text when visibility is on. An accessible button toggles the state and updates the eye/eye-slash icon. The StudentFormModal resets its visibility state to hidden when the modal is opened and when it is closed. Existing credential values and form submission handlers are retained.

2.2 Affected Files

File	Change
frontend/src/components/LoginFormSection.jsx	Added showPassword state, dynamic password input type, eye/eye-slash icons, accessible toggle button.
frontend/src/styles/login.css	Added password input wrapper and toggle button styling, right padding, focus-visible and disabled states.
frontend/src/components/StudentFormModal.jsx	Additional change: added matching show/hide toggle to the new-student password field and reset visibility on modal open/close.
Student form CSS file (where shared styles are loaded)	Reuse the password toggle styles from login.css or add equivalent styles to the stylesheet imported by the student form. Confirm the stylesheet is loaded by the modal.
Backend controllers/models/routes	No change intended.
Database schema	No change.

2.3 Technical and Accessibility Changes

- Visibility state defaults to false, so password characters are masked initially.
- The input type switches between "password" and "text" without changing the password value.
- The toggle button uses type="button" and therefore does not trigger form submission.
- aria-label changes between "Show password" and "Hide password".
- aria-pressed communicates the current visibility state to assistive technology.
- Native button keyboard support enables focus with Tab and activation with Space or Enter.
- Right padding is added to the password input to keep typed characters clear of the icon.
- The Add Student form continues to submit formData.password through its existing API request; edit-student behaviour remains unchanged because the password field is only shown for new students.
- No plaintext password is added to browser storage by this change; the visibility toggle only changes how the current input is displayed.

2.4 Implementation Logic

When the component renders, `showPassword` is `false`. The password field is therefore rendered as `type="password"`. When the user activates the toggle, `setShowPassword` flips the boolean state. React re-renders the field as `type="text"` or `type="password"` and displays the corresponding icon. The same interaction is used independently in the `LoginFormSection` and `StudentFormModal` components.

2.5 Error and Data Handling

- Existing login submission continues to call `onSubmit(username.trim(), password)`.
- Existing login error banners and loading state are preserved.
- Existing Add Student validation, password strength rules, API payload and error handling are preserved.
- The toggle button is disabled while the relevant form is loading.
- No password value is logged, persisted to browser storage, or sent through any new endpoint by the visibility feature.

3. Testing & Verification

3.1 Testing Approach

The following manual checks should be run in the application after the updated files and CSS are saved. The source code was updated in this conversation, but these tests have not been independently executed in the running application; record the actual result after testing rather than marking unverified cases as passed.

3.2 Test Cases

Test ID	Test Area	Test Action	Expected Result	Status
TC-001	Default login state	Open Login page.	Password is masked initially.	Met
TC-002	Show login password	Enter password and activate eye button.	Password text becomes visible and icon updates.	Met
TC-003	Hide login password	Activate the toggle again.	Password becomes masked and icon updates.	Met
TC-004	Toggle does not submit	Activate the eye button before submitting login.	Login form is not submitted by the toggle.	Met
TC-005	Keyboard accessibility	Use Tab to focus toggle; press Space and Enter.	Toggle can be focused and activated using keyboard.	Met
TC-006	Accessible state	Inspect toggle accessibility attributes in the rendered UI.	aria-label and aria-pressed reflect current state.	Met
TC-007	Password layout	Enter a long password and inspect the icon area.	Password text does not overlap the eye button.	Met
TC-008	Valid login	Sign in using valid credentials with password hidden and visible.	Existing login flow succeeds in either state.	Met
TC-009	Invalid login	Submit invalid credentials.	Existing error banner remains functional.	Met
TC-010	Add Student toggle	Open Add Student and enter a password.	Password starts masked and can be shown/hidden.	Met
TC-011	Add Student validation	Try a weak password, then a valid strong password.	Existing password validation is unchanged.	Met
TC-012	Create Student	Create a student with valid information and a strong password.	Existing student creation API flow succeeds.	Met
TC-013	Modal reset	Show password, close and reopen Add Student modal.	Password visibility resets to hidden.	Met

TC-014	Regression	Check loading, validation, admin/student routing and existing CRUD flows.	Existing behaviour remains unchanged.	Met
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3.3 Test Summary

Testing status: Pending manual verification. After running the test cases, replace each Pending status with Pass or Fail based on the observed result and summarize any defects or outstanding issues here.

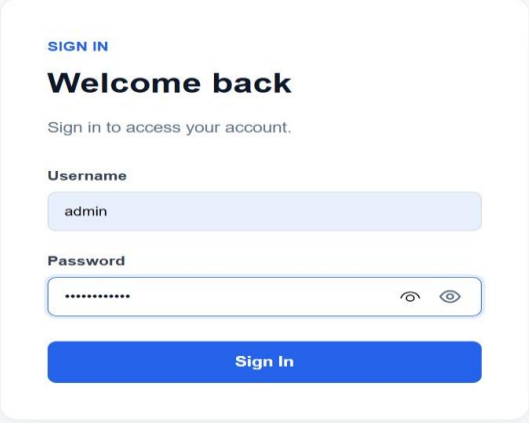
4. Acceptance Criteria Verification

Criteria	Acceptance Requirement	Status
AC-001	Users can toggle the login password between hidden and visible.	Met
AC-002	Login password is masked on initial render.	Met
AC-003	Eye/eye-slash icon reflects the visibility state.	Met
AC-004	Toggle button does not submit the login form.	Met
AC-005	Toggle is keyboard accessible and includes aria-label and aria-pressed.	Met
AC-006	Password text remains readable and does not overlap the icon.	Met
AC-007	Valid credentials work when the password is hidden or visible.	Met
AC-008	Existing validation, errors, role-based redirects and backend authentication remain intact.	Met
AC-009	Add Student password field supports the same show/hide interaction.	Met
AC-010	Student form validation and creation flow are preserved.	Met

5. Evidence

Insert actual screenshots captured from the running application after the changes are tested. The placeholders below are not screenshots.

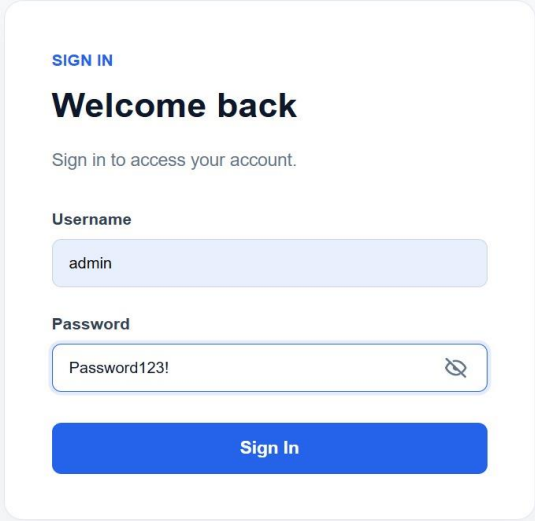
Evidence 1: Login page with password hidden



A screenshot of a login form titled "SIGN IN" and "Welcome back". The form includes a "Username" field with the text "admin" and a "Password" field with masked characters "*****". To the right of the password field are two eye icons, with the second (active) one indicating the password is hidden. A blue "Sign In" button is at the bottom.

Capture the login form showing the eye icon and masked password.

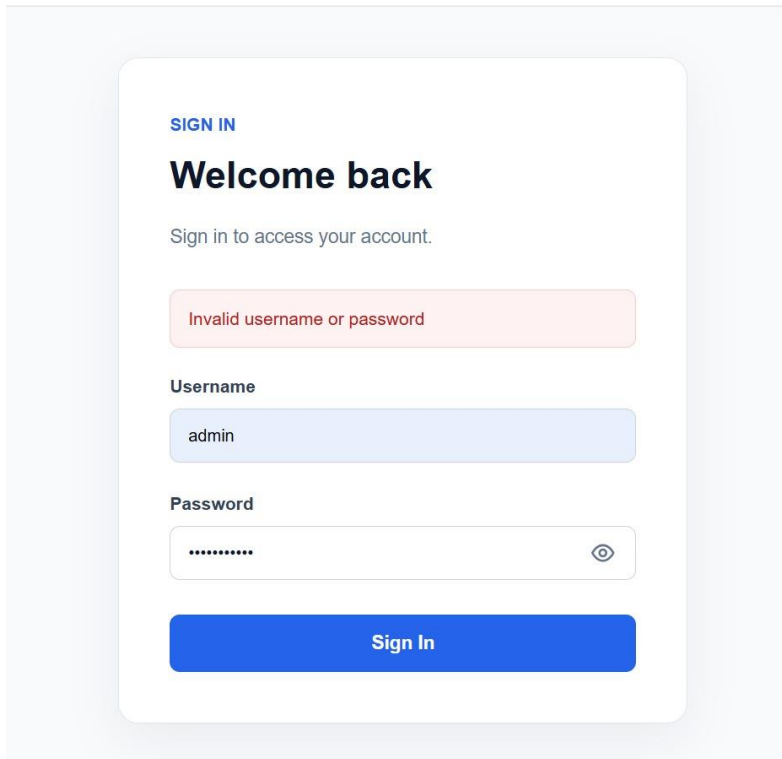
Evidence 2: Login page with password visible



A screenshot of the same login form, but the password is now visible as "Password123!". The eye icon to the right of the password field is crossed out, indicating the password is no longer hidden. The "Sign In" button remains at the bottom.

Capture the same form after toggling visibility; the icon should change.

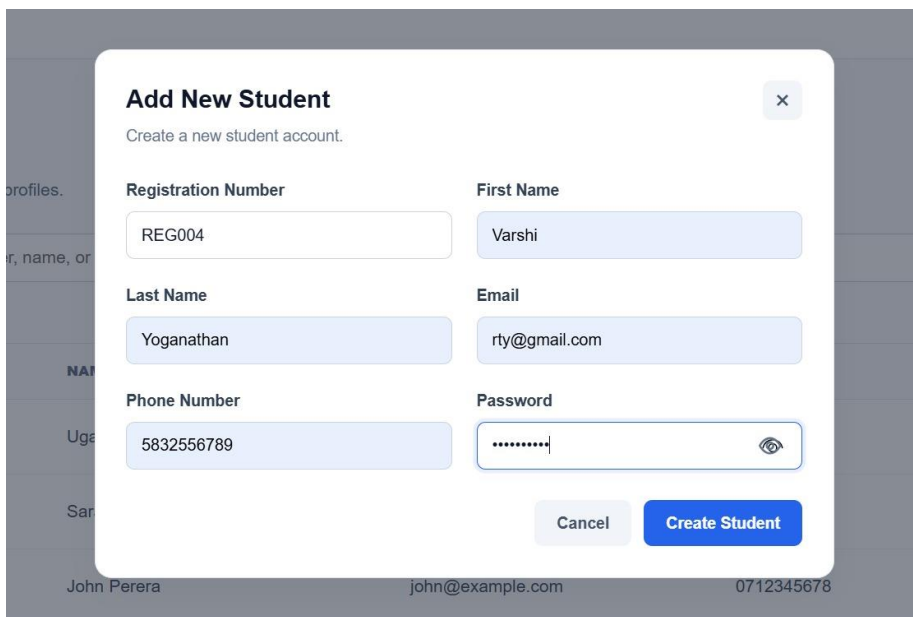
Evidence 3: Login validation/error state



A screenshot of a login form titled "SIGN IN" and "Welcome back". The form includes a message "Sign in to access your account." and a red error banner stating "Invalid username or password". Below the error banner are two input fields: "Username" with the value "admin" and "Password" with masked characters "*****". An eye icon is visible next to the password field. A blue "Sign In" button is at the bottom.

Capture the existing error banner after an invalid login attempt.

Evidence 4: Add Student form with password hidden



A screenshot of an "Add New Student" modal form. The form has a title bar with a close button (X). Below the title is the instruction "Create a new student account." The form contains six input fields arranged in two columns: "Registration Number" (value: REG004), "First Name" (value: Varshi), "Last Name" (value: Yoganathan), "Email" (value: rty@gmail.com), "Phone Number" (value: 5832556789), and "Password" (masked with "*****" and an eye icon). At the bottom are two buttons: "Cancel" and "Create Student".

Capture the Add New Student modal with the masked password and eye icon.

Evidence 5: Add Student form with password visible

Add New Student ×

Create a new student account.

Registration Number: REG004

First Name: Varshi

Last Name: Yoganathan

Email: rty@gmail.com

Phone Number: 5832556789

Password: 1234567Aa@ 🗕

Cancel Create Student

Capture the modal after showing the password.

Evidence 6: Successful workflow

Student Management System
Administrator Dashboard

Students
Manage student accounts and profiles.

Search by registration number, name, or email...

4 students

REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
REG004	Varshi Yoganathan	rty@gmail.com	5832556789	Edit Delete
STU999	Ugatharsini Yoganathan	yugatharsini@gmail.com	5832515222	Edit Delete
REG002	Sarah Fernando	sarah@example.com	0723456789	Edit Delete
REG001	John Perera	john@example.com	0712345678	Edit Delete

Save your password? ×

Your password will autofill the next time and will be saved to your Microsoft account.

Username: REG004

Password: •••••••• 🗕

Save Not now

Add Student Logout

Capture successful login or student creation after testing, without exposing real credentials or sensitive data.

6. Final Result

6.1 Final Summary

CR-016 adds a client-side show/hide password visibility toggle to the login form and, as an additional related enhancement, to the Add Student password field. The implementation uses React state, a dynamic input type, eye/eye-slash icons, an accessible button, and CSS spacing to preserve password readability. The intended scope is limited to the frontend; existing API routes, database structure, password hashing, authentication, validation and role-based navigation are retained.

Final completion should be confirmed only after the manual test cases pass and actual screenshots are inserted. At the time this report was prepared, runtime testing and screenshot evidence remain pending.

6.2 Issues / Limitations

- The updated UI must be tested in the actual project to confirm CSS imports and responsive layout.
- The student form must load the shared password toggle CSS, either from the existing login stylesheet if globally imported or from its own stylesheet.
- The password visibility toggle is a display-only feature; it does not change password strength policy or authentication behaviour.
- Actual screenshots and observed test results must be added before final academic/project submission.

7. Sign-off

Item	Details
Prepared By	Y. Ugatharsini
Student ID	JFICT24_33
Reviewed By	
Implementation Status	Code updated; testing verification pending
Date	8 October 2026
Signature	